

Bi-weekly Report

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Group No: 14

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Project title Production of Aniline by Vapour-Phase Catalytic Hydrogenation of Nitrobenzene

Team members

Name	Roll No.	Name	Roll No.
Nithin T M	230709	Shubham Singh	230999
Aditya Nitin Patil	230074	Shivam Dongare	230971
Nithin D H	230708	Sajag Masane	230896

Objectives for this report

1. Process identification and selection of the synthesis route.
2. Specification of the product and the feed streams.
3. Preparation of the block diagram.
4. Input–output cost analysis and preliminary economic feasibility test.
5. Selection and validation of the thermodynamic model.
6. Tuning of the thermodynamic model parameters by regression.

1 Process Identification and Route Selection

1.1 Why aniline has been chosen

Aniline ($\text{C}_6\text{H}_5\text{NH}_2$) is a bulk aromatic amine and it is one of the most important intermediates of the organic chemical industry; about 85 per cent of the world's production is used in the manufacture of MDI (methylene diphenyl diisocyanate), which is further used for rigid polyurethane foam, insulation panels and adhesives. The rest is allocated to rubber processing chemicals, agrochemicals, dyes and pharmaceutical intermediates.

In 2026 the global aniline market had a value of about USD 8.2 billion and it is projected to reach approximately USD 12.2 billion by 2033, with a CAGR of nearly 5.8 %. Asia–Pacific is holding the largest share, close to 39 % of the market, which makes an Indian production facility a reasonable suggestion; India also possesses an existing nitrobenzene manufacturing base, so the feedstock is available domestically.

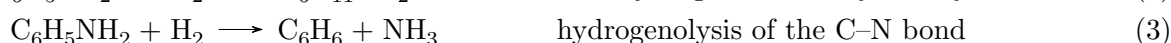
1.2 The chemistry

The route selected is the vapour-phase catalytic hydrogenation of nitrobenzene over a supported copper or nickel catalyst. The nitro group is reduced in stages to the amine group and water is released as the only main by-product.

Main reaction



Side reactions



Both the side reactions are proceeding through aniline itself, therefore the overall consumption of nitrobenzene for each side product is one mole, and the water formation stays at two moles per mole of nitrobenzene reacted in each instance; this point was later made use of when closing the material balance.

The reaction mainly releases a large amount of heat. This is the most important feature of the process from the standpoint of design. Our capacity yields a heat release amounting to 17.6 MW (computed in Section 3.5), which is enough to raise a significant quantity of steam, and at the same time it creates a genuine hot-spot and runaway risk within the reactor. These points will be dealt with in the later reports.

1.3 Comparison of the available routes

Route	Catalyst / medium	Selectivity	Remarks
Vapour-phase (<i>selected</i>)	Cu or Ni on support, H ₂ gas	95–98 %	Continuous, clean by-products, heat recoverable. Hydrogen handling and catalyst coking are the concerns.
Liquid-phase	Pd/C, Pt or Raney Ni in solvent	95–97 %	Easier temperature control, but solvent handling, catalyst separation and more side products.
Béchamp	Iron filings + HCl	85–95 %	Cheap reagents but massive Fe ₃ O ₄ sludge; largely superseded.
Phenol amination	NH ₃ at high pressure	—	Economical only where phenol is cheap; not suited to Indian conditions.

We went with the vapour-phase route. It runs continuously, gives the highest selectivity of the four options, produces only water as the bulk by-product, and the reaction heat can be recovered as steam rather than just dumped.

The battery limits start from purchased nitrobenzene, so nitration of benzene is not part of this project. This keeps the mixed acid system, the electrolyte thermodynamics and the spent acid reconcentration outside our scope. Section 5 discusses what the decision costs us economically.

2 Feed and Product Specifications

Stream	Composition / purity	Temp.	Pressure
Aniline product	≥ 99.9 wt%; water < 0.1 wt%; nitrobenzene < 5 ppm	40 °C	1.0 bar
Nitrobenzene feed	≥ 99.5 wt%, technical grade	Ambient	1.0 bar
Hydrogen make-up	99.9 mol%, balance N ₂	Ambient	5 bar
Treated wastewater	Aniline < 1 ppm before discharge	40 °C	1.0 bar

The 5 ppm nitrobenzene limit in the product is what really controls the design here — nitrobenzene is toxic and it colours the product besides. This limit is what will drive the aniline column design in Report 3.

3 Design Basis and Material Balance

3.1 Basis

Plant capacity	100,000 t/y of aniline
Operating hours	8000 h/y
Production rate	12.5 t/h
Nitrobenzene conversion per pass, X	99 %
Selectivity to aniline, S_{AN}	95 %
Selectivity to cyclohexylamine	1.5 %
Selectivity to benzene (and ammonia)	3.5 %
Hydrogen to nitrobenzene ratio at reactor inlet	10:1 molar
Recycle gas purity (assumed)	95 mol% H ₂ , balance N ₂

Definition of selectivity used throughout this report. All three selectivities are defined on a *moles of nitrobenzene reacted* basis, that is

$$S_i = \frac{\text{moles of nitrobenzene converted into product } i}{\text{total moles of nitrobenzene reacted}}$$

and by definition they sum to 100 %: $95 + 1.5 + 3.5 = 100$. Since every molecule of nitrobenzene carries exactly one benzene ring and one nitrogen atom, and since both side products are formed from aniline itself, this definition coincides with a carbon basis and with a product-mole basis for this particular reaction network. The three definitions are therefore numerically identical here, but the reacted nitrobenzene basis is the one we have used and it should be quoted as such.

3.2 Equations used

Let F be the nitrobenzene entering the reactor, X the conversion per pass and S the selectivity. Then:

$$\text{Aniline produced} = F X S_{AN} \quad (4)$$

$$\text{Nitrobenzene reacted} = F X \quad (5)$$

$$\text{Unreacted nitrobenzene (recycled)} = F (1 - X) \quad (6)$$

$$\text{Water produced} = 2 F X \quad (7)$$

$$\text{Hydrogen consumed} = 3 (\text{aniline}) + 6 (\text{cyclohexylamine}) + 4 (\text{benzene}) \quad (8)$$

Because the unreacted nitrobenzene goes back to the reactor inlet, the fresh make-up equals what's actually consumed by reaction — not the total flow entering the reactor. We've kept that distinction clear in the table below.

3.3 Overall material balance

Stream / quantity	kmol/h	kg/h
Aniline product	134.22	12 500
Nitrobenzene to reactor inlet	142.71	17 571
Nitrobenzene reacted	141.28	17 395
Unreacted nitrobenzene (recycle)	1.43	176
Fresh nitrobenzene make-up	141.28	17 395
Cyclohexylamine formed	2.12	210
Benzene formed	4.94	386
Ammonia formed	4.94	84
Water formed	282.56	5 091
Hydrogen consumed	435.15	877
Hydrogen at reactor inlet (10:1)	1427.1	2 877
Hydrogen recycled	991.9	2 000

Carbon closes exactly here: $134.22 + 2.12 + 4.94 = 141.28$ kmol/h, matching the nitrobenzene reacted.

3.4 Specific consumption figures

Nitrobenzene per tonne of aniline	1.392 t
Hydrogen consumed per tonne of aniline	70.2 kg
Hydrogen purchased per tonne of aniline	71.6 kg
Water produced per tonne of aniline	0.407 t
Benzene formed per tonne of aniline	30.9 kg
Ammonia formed per tonne of aniline	6.7 kg
Cyclohexylamine formed per tonne of aniline	16.8 kg

3.5 Hydrogen loop — make-up, recycle and purge

The hydrogen figures in the table above describe what happens across the reactor only. Since a purge is unavoidable, the make-up requirement is not the same as the consumption, and the closure of the loop is worked out here explicitly.

Nitrogen enters at 0.1 mol% in the make-up and it does not react. At steady state it has to leave at the same rate, and the purge is the only exit available. Taking the recycle gas purity as 95 mol% H_2 , the two balances are:

$$\text{Nitrogen: } 0.001 M = y_{\text{N}_2} P \quad (9)$$

$$\text{Hydrogen: } 0.999 M = 435.15 + y_{\text{H}_2} P \quad (10)$$

where M is the fresh make-up and P the purge, both in kmol/h. Solving the two together gives $P = 0.02 M$ and hence:

Quantity	kmol/h	kg/h
Fresh H_2 make-up, 99.9 mol%	444.0	895
Hydrogen consumed by reaction	435.2	877
Hydrogen lost in the purge	8.4	17
Gas leaving the flash separator	1044.2	—
of which nitrogen	52.2	—
Purge stream	8.9	—
Recycle gas returned to the reactor	1035.3	—
Check: H_2 at reactor inlet	1427.1	2 877

The loop therefore closes: $1035.3 \times 0.95 + 444.0 \times 0.999 = 1427.1$ kmol/h, which is exactly the 10:1 ratio on the 142.71 kmol/h of nitrobenzene entering the reactor. The purge is only about 2 % of the make-up and it carries away roughly 1.9 % of the hydrogen purchased.

Purge loss. On a purchased basis the hydrogen requirement is 71.6 kg per tonne of aniline against 70.2 kg/t of pure reaction consumption, so about 2 % of the hydrogen bought leaves with the purge without ever reacting. In the cost analysis of Section 5 we have used the consumption figure; using the purchase figure instead would add about Rs. 259 per tonne to the raw material cost. This shall be carried properly once the recycle loop is converged in Aspen.

The assumed recycle purity of 95 mol% is itself a design variable. A tighter purge raises the hydrogen recovery but allows nitrogen to build up, which lowers the hydrogen partial pressure at the reactor inlet and hurts the rate. The optimum is to be found by a sensitivity study in Report 4.

3.6 Reaction heat duty

The heat release has been computed from the actual reaction extents rather than quoted as a round figure. All the nitrobenzene reacted passes first through the main reaction, and the two side products are then formed from aniline, so the extents are:

Reaction	Extent (kmol/h)	ΔH_r (kJ/mol)	Duty (kW)
$\text{C}_6\text{H}_5\text{NO}_2 + 3\text{H}_2 \longrightarrow \text{C}_6\text{H}_5\text{NH}_2 + 2\text{H}_2\text{O}$	141.28	-443	17 385
$\text{C}_6\text{H}_5\text{NH}_2 + 3\text{H}_2 \longrightarrow \text{C}_6\text{H}_{11}\text{NH}_2$	2.12	≈ -193	113
$\text{C}_6\text{H}_5\text{NH}_2 + \text{H}_2 \longrightarrow \text{C}_6\text{H}_6 + \text{NH}_3$	4.94	≈ -51	69
Total heat of reaction			17 568

That is **17.6 MW**, and not 16.5 MW as would be obtained by taking the aniline production rate of 134.22 kmol/h in place of the nitrobenzene reacted. The distinction matters because every mole of nitrobenzene converted releases the main reaction heat, including the 7.06 kmol/h which subsequently goes on to form side products. The side reactions themselves contribute only about 0.18 MW, which is 1 % of the total, but the correct main-reaction extent is 5 % higher than the aniline rate and that is not negligible when sizing the steam generation.

The heats of the two side reactions have been estimated from standard gas-phase enthalpies of formation and are indicative; they shall be replaced by Aspen values once the reactor is set up with real kinetics.

4 Block Diagram

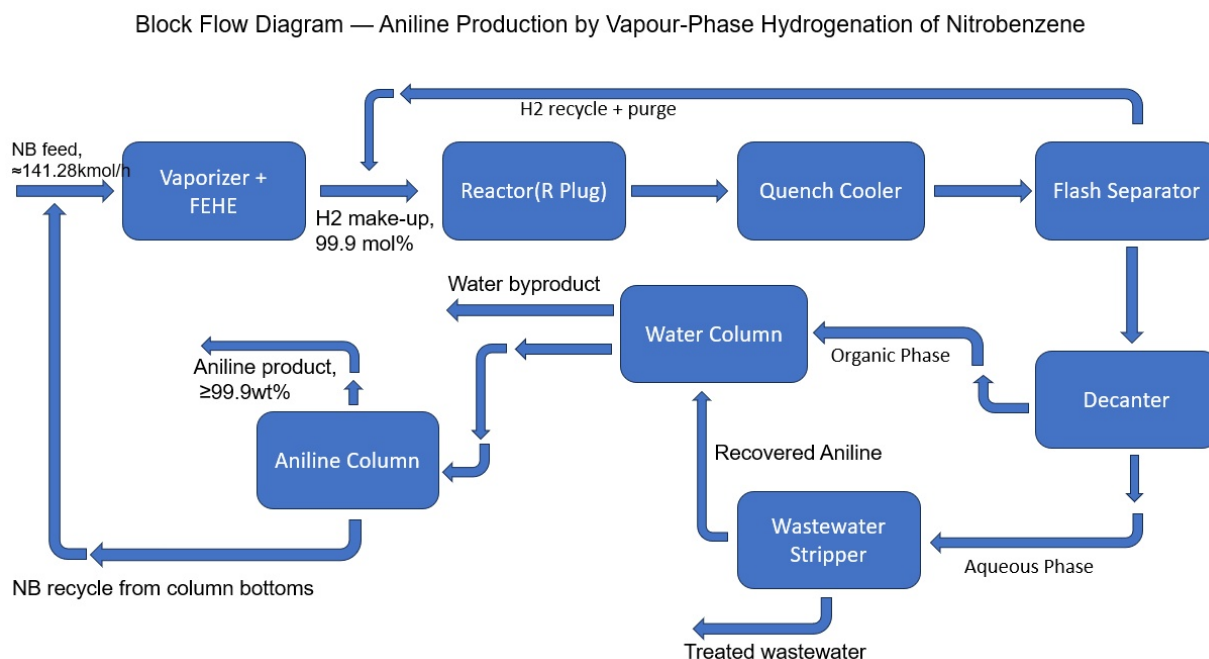


Figure 1: Block flow diagram of the aniline plant, with the three recycle streams labelled.

The reason for the sequence is this: nitrobenzene is vapourised and superheated against the reactor effluent, mixed with make-up and recycled hydrogen, and then passed over the catalyst bed. The effluent is cooled in two stages and the excess hydrogen is flashed off, compressed and returned, with a purge to prevent the buildup of nitrogen and the by-products from light. The liquid which has been condensed divides into two liquid phases in the decanter, which is possible only because aniline and water are partially miscible — this is precisely the property dealt with in Section 6 on thermodynamics. The organic phase is dried in the water column and purified in the aniline column, the bottom portion of which sends nitrobenzene back to the feed. The aniline which is dissolved is removed by means of the aqueous phase before the effluent is treated.

5 Input–Output Cost Analysis and Preliminary Economic Feasibility

5.1 Purpose and approach

Input–output analysis is the crudest economic screen there is, and also the first one worth doing. In its strictest form you only look at the raw materials against the main product. The logic is blunt: if the product’s value can’t even cover raw material cost, no clever design downstream is going to rescue the project, so it should be dropped right here.

Raw material cost dominates this process, so we didn’t stop at the crude version — non-feed operating costs and a limited credit for recoverable by-products and heat were added in too. We’ve laid the result out as a *margin improvement pathway*, so each commercial lever’s contribution is visible on its own rather than the whole result hanging on one convenient assumption.

5.2 Material consumption basis

Nitrobenzene, with recycle

$$\text{Fresh NB factor} = \frac{M_{\text{NB}}}{M_{\text{AN}}} \cdot \frac{1}{S_{\text{AN}}} = \frac{123.11}{93.13} \cdot \frac{1}{0.95} = 1.392 \text{ t/t} \quad (11)$$

Notice conversion doesn’t show up in this factor. That’s because the flowsheet recycles the unconverted nitrobenzene from the aniline column bottoms straight back to the reactor inlet. So what

the plant actually buys fresh is set by how much reacts, which comes down to selectivity alone. The 99 % conversion figure decides reactor size, recycle flow, energy demand — all of that — but it doesn't mean the leftover 1 % needs buying again. Dividing by conversion *and* selectivity would count the recycled material as if it were lost, and overstate what we actually need to purchase.

Hydrogen

We corrected the hydrogen factor on the same 95 % selectivity basis, folding in the hydrogen consumed by the two side pathways too — 6 mol per mol of cyclohexylamine and 4 mol per mol of benzene. That works out to **0.0702 t/t**. Sticking with the simple main-reaction figure of 0.065 t/t would have understated hydrogen consumption by roughly 8 %.

5.3 Commercial price and procurement basis

Every price below is tied to a specific engineering or procurement strategy — none of them were picked backward from a margin we wanted to show.

Item	Rs./t	Basis / rationale	Reference
Aniline	155,000	Sustained realisation target, equivalent to 1631 USD/t at Rs. 95 per USD. The Indian quoted band was 1500–1600 USD/MT, so this sits marginally above the top of the observed range and requires product quality and market access to be maintained.	Ref. 8(a); FX rate as on 6 August 2026
Nitrobenzene	85,500	Bulk / term procurement target, 900 USD/t, about 6 % below a spot reference of Rs. 91,200/t (960 USD/t). A plant consuming 139,000 t/y is a large and predictable buyer, which gives real negotiating leverage.	Ref. 8(b) for the spot reference
Hydrogen	185,000	On-site SMR <i>production</i> cost of 1.95 USD/kg, not a merchant purchase price. Published Indian grey hydrogen production cost is 1.92–2.43 USD/kg at delivered gas of 9–12 USD/MMBtu; we are at the lower end because the site already has natural gas for the gas turbine and the reformer heat can be recovered.	Ref. 8(d)
Benzene	90,250	Used only for the back-integration screening figure of Rs. 9,500/t of aniline; 0.95 USD/kg, with a stoichiometric requirement of 0.634 t benzene per t of nitrobenzene.	Ref. 8(c)
Electricity	embedded	Supplied from the captive gas turbine; included within the operating cost basis.	—

Procurement strategy. We'd want at least two qualified nitrobenzene suppliers on the books, and domestic supply needs to be checked against import parity on a *delivered* basis — freight, insurance, port handling, duties, any trade-remedy measures included — not just FOB price. A benzene-linked formula contract, if one can be arranged, won't raise the average margin by itself, but it cuts exposure to sudden spot spikes and makes financing easier to secure.

5.4 Raw material cost and gross contribution

Item	Rs./t aniline	Rs. crore/y
Aniline revenue	155,000	1550.0
Nitrobenzene, $1.392 \times 85,500$	118,972	1189.7
Hydrogen, $0.0702 \times 185,000$	12,987	129.9
Total raw materials	131,959	1319.6
Gross contribution	23,041	230.4
Gross margin	14.9 %	

5.5 Comment on the result

The gross contribution of Rs. 23,041 per tonne, that is 14.9 % of the sales revenue, is enough to clear the input–output screen and the project therefore goes forward.

The table also shows where the money goes. Nitrobenzene alone takes about 77 % of the sales revenue, so procurement is the single most important commercial function in this plant. Since the consumption factor multiplies any price movement, every Rs. 1,000/t rise in the nitrobenzene price adds Rs. 1,392/t to the production cost; the corresponding figure for hydrogen is Rs. 702/t for every Rs. 10,000/t.

5.6 Limitations

- Utilities, labour, maintenance and capital charges are not included, so this is a raw material screen only and not a profitability estimate.
- The Rs. 185,000/t hydrogen cost assumes an on-site reformer and needs an SMR mass and energy balance to confirm it.
- Equipment costs and working capital are not estimated, so ROCE, NPV, IRR and payback are taken up in Report 5.

The remaining weaknesses are collected in Section 9.

6 Thermodynamic Model Selection and Validation

6.1 Why an activity coefficient model is required

The mixture handled in this plant is strongly non-ideal. Aniline and water are only partially miscible whereas aniline and nitrobenzene are completely miscible, and a cubic equation of state cannot represent liquid phase splitting of such a polar–non-polar system. An activity coefficient model is therefore needed for the liquid phase, with an equation of state for the vapour.

NRTL-RK has been selected. NRTL handles the liquid phase activity coefficients and Redlich–Kwong the vapour. NRTL is preferred over Wilson because the Wilson equation cannot predict liquid–liquid splitting at all, and over UNIQUAC because sufficient binary data is available to fit NRTL directly for this system.

Of the phase equilibria relevant here, VLE arises in the columns, LLE in the decanter and VLLE at the heterogeneous azeotrope where the two liquid phases boil together. SLE does not arise, as nothing crystallises in the operating range.

Hydrogen and nitrogen have been declared as **Henry components**. They are supercritical at the operating conditions and so have no pure component vapour pressure; their solubility must be described by Henry’s law and not Raoult’s law. If this is not done the dissolved hydrogen in the liquid streams is computed wrongly and the overall mass balance does not close.

6.2 The problem with the databank parameters

Before touching anything else, we first checked the binary interaction parameters Aspen Plus pulls in automatically.

	Component i	Component j	Source	Temp. Units	AU	AJ	BU	BJ	CU	DU	EU	EJ	FU	FJ	TLOWER	TUPPER
	ANILINE	NITRO-01	APV140 VLE-RK	C	0	0	-217.928	287.592	0.3	0	0	0	0	0	83	99
	ANILINE	WATER	APV140 VLE-RK	C	-0.775	2.2118	465.474	364.09	0.3	0	0	0	0	0	99	168
	ANILINE	BENZE-01	APV140 VLE-RK	C	1.9429	-2.8485	-717.708	1302.27	0.3	0	0	0	0	0	25	180.5
	ANILINE	CYCLO-01	APV140 VLE-RK	C	3.0918	-2.1102	-910.317	617.869	0.3	0	0	0	0	0	70.89	181.45
	NITRO-01	WATER	APV140 LLE-AS...	C	-5.1549	5.8547	2270.62	229.497	0.2	0	0	0	0	0	0	106.6
	NITRO-01	BENZE-01	APV140 VLE-RK	C	0.3005	1.446	40.2665	-653.816	0.3	0	0	0	0	0	70	211
	WATER	BENZE-01	APV140 LLE-AS...	C	140.087	45.1905	-5954.31	591.368	0.2	0	-20.0254	-7.5629	0	0	0.8	77
	BENZE-01	CYCLO-01	APV140 VLE-RK	C	-0.8986	0.3579	676.613	-388.303	0.3	0	0	0	0	0	60.62	133.77

Figure 2: NRTL binary interaction parameters as retrieved from the Aspen Plus databanks, before regression.

Two things stood out here, and together they’re the reason we went ahead with a regression of our own.

1. The ANILINE–WATER row comes from **APV140 VLE-RK** — a *vapour–liquid* data bank — even though this pair is really a liquid–liquid splitting pair. Tellingly, Aspen itself pulled the NITRO-01–WATER and WATER–BENZE-01 pairs from **LLE-ASPEN** with $C_{ij} = 0.2$, which is the correct setting for liquid–liquid systems. So the software treats similar pairs differently, and not in the right direction for the one we care most about.
2. The stated validity range for ANILINE–WATER is $T_{\text{LOWER}} = 99^\circ\text{C}$ to $T_{\text{UPPER}} = 168^\circ\text{C}$. Our decanter runs at around 40°C — nearly 60 degrees below where this correlation is actually valid. Aspen will happily extrapolate and hand you a number, but there’s no experimental data backing it up down there.

Given that the whole decanter design, and by extension the entire separation sequence, rides on how well this one pair is represented, we couldn’t just accept the default parameters as they stood.

6.3 Experimental data used

We pulled the data through the NIST TDE facility built into Aspen Plus. Aniline + water happens to be one of the recommended reference systems for liquid–liquid equilibrium studies — several independent workers have reported consistent numbers for it, so the data quality here is genuinely good.

TDE handed back 19 LLE sets, 23 VLE sets, one azeotropic set and some diffusion coefficient data. We didn’t use all of it — overlapping sets from different authors mostly just add scatter, not information. Here’s what we picked:

Data set	Type	Purpose
BLLE001, 002, 005, 019	Binary LLE	Used for the regression
AZEO001	Azeotropic data	Kept aside for validation, not used in the fit
BVLE004, 023, 025	Isobaric VLE	Retained for a later check

We dropped the diffusion coefficient data — it’s a transport property, and has nothing to do with phase equilibrium anyway.

6.4 Regression procedure and the over-parameterisation problem

We set the regression up with the maximum likelihood objective function and the Britt–Luecke algorithm, using Deming initialisation. The non-randomness parameter C_{ij} was fixed at 0.2 — the standard value for liquid–liquid systems — rather than regressing it alongside the others.

Our first attempt regressed all four parameters, a_{ij} , a_{ji} , b_{ij} and b_{ji} . Here’s what came out.

Parameter	Comp. i	Comp. j	Value	Std. deviation
NRTL/1	ANILINE	WATER	−4.579	0.487
NRTL/1	WATER	ANILINE	4.863	0.636
NRTL/2	ANILINE	WATER	1419.39	160.79
NRTL/2	WATER	ANILINE	20.9987	208.37

Look at that last parameter: a standard deviation of 208 against a value of just 21. The data simply isn't pinning this parameter down at all — it's floating free, and Aspen flagged a warning that one parameter had hit its bound. Four parameters is one too many for LLE data of this kind.

So we reran it with b_{ji} fixed at zero, fitting only three parameters this time.

Parameter	Comp. i	Comp. j	Value	Std. deviation
NRTL/1	ANILINE	WATER	−4.62238	0.2216
NRTL/1	WATER	ANILINE	4.92646	0.0577
NRTL/2	ANILINE	WATER	1433.47	74.48

The parameter values barely moved, but every standard deviation dropped by somewhere between two and tenfold. That settles it — the three-parameter fit is the one we're keeping.

	Component i	Component j	Source	Temp. Units	ALJ	BLJ	BU	BJL	CU	DU	EU	EJL	FU	FJL	TLOWER	TUPPER
▶	ANILINE	NITRO-01	APV140 VLE-RK	C	0	0	-217.928	287.592	0.3	0	0	0	0	0	83	99
▶	ANILINE	WATER	R-DR-1	C	-4.62238	4.92646	1433.47	0	0.2	0	0	0	0	0	99	168
▶	ANILINE	BENZE-01	APV140 VLE-RK	C	1.9429	-2.8485	-717.708	1302.27	0.3	0	0	0	0	0	25	180.5
▶	ANILINE	CYCLO-01	APV140 VLE-RK	C	3.0918	-2.1102	-910.317	617.869	0.3	0	0	0	0	0	70.89	181.45
▶	NITRO-01	WATER	APV140 LLE-AS...	C	-5.1549	5.8547	2270.62	229.497	0.2	0	0	0	0	0	0	106.6
▶	NITRO-01	BENZE-01	APV140 VLE-RK	C	0.3005	1.446	40.2665	-653.816	0.3	0	0	0	0	0	70	211
▶	WATER	BENZE-01	APV140 LLE-AS...	C	140.087	45.1905	-5954.31	591.368	0.2	0	-20.0254	-7.5629	0	0	0.8	77
▶	BENZE-01	CYCLO-01	APV140 VLE-RK	C	-0.8986	0.3579	676.613	-388.303	0.3	0	0	0	0	0	60.62	133.77
▶																

Figure 3: NRTL parameters after regression. The ANILINE–WATER row now shows the source as R-DR-1, that is our own regression case, with C_{ij} corrected to 0.2.

6.5 Quality of the fit

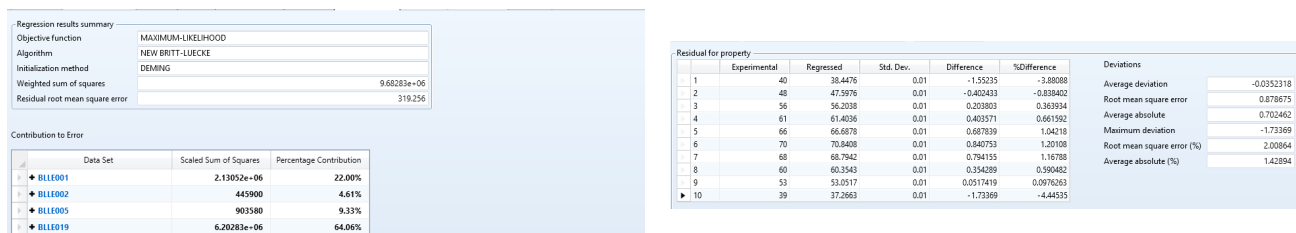


Figure 4: Left: regression summary and the contribution of each data set to the total error. Right: residuals in temperature for BLLE019, the set nearest the decanter operating range.

The regression used the maximum likelihood objective function with the Britt–Luecke algorithm and Deming initialisation, which is the Aspen default for this kind of work. The weighted sum of squares came to 9.68×10^6 with a residual root mean square error of 319.3.

The error is not spread evenly across the four data sets. BLLE019 alone contributes 64 % of the total while BLLE002 contributes only 4.6 %, BLLE001 22 % and BLLE005 9.3 %. This tells us the four sets are not of equal quality, and it is worth knowing which of them the fit is really being pulled by.

Taking the residuals in temperature, BLLE019 gives a root mean square error of 0.879 °C and an average absolute deviation of 0.703 °C, that is about 1.4 % in relative terms. BLLE002 does better at 0.559 °C and 0.554 °C, roughly 1.0 %. We have quoted BLLE019 in particular because it covers 39–70 °C, which is the range the decanter will actually operate in. Both figures are well within what we would accept for design purposes.

6.6 Effect on the phase behaviour

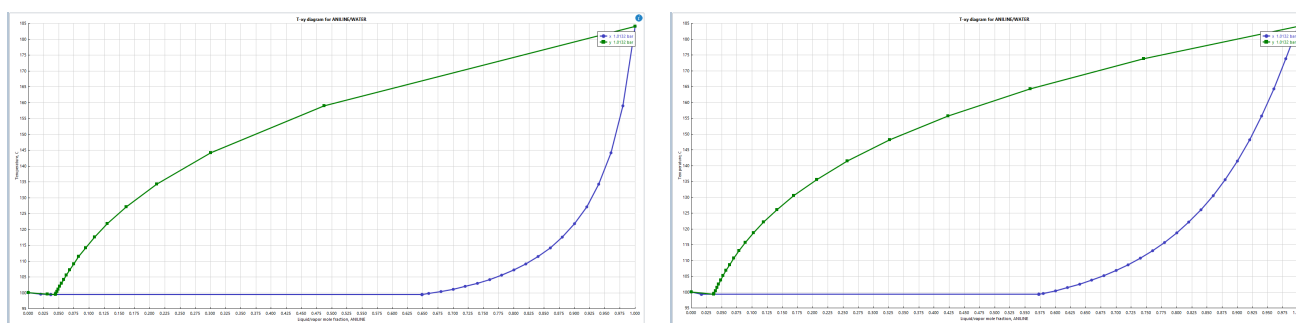


Figure 5: T - xy diagram for aniline–water, before regression (left) and after (right). The flat portion is the two-liquid region.

That flat horizontal stretch in both diagrams is the signature of a **heterogeneous azeotrope**. Two liquid phases coexist across this composition range, so the mixture boils at a fixed temperature and gives off a vapour of fixed composition. This is exactly why the decanter works in our flowsheet — and exactly why a plain distillation column, on its own, can’t separate aniline from water.

Regression narrowed the miscibility gap on both sides. Before, it ran from about $x_{\text{aniline}} = 0.05$ to 0.65; afterwards it’s tighter, from about 0.037 to 0.575.

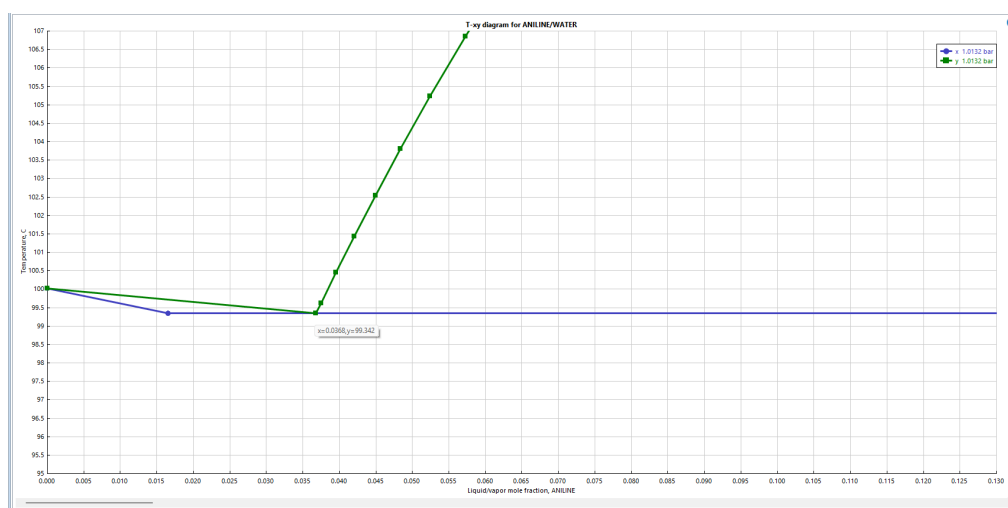


Figure 6: Magnified view near the azeotropic point after regression. The calculated azeotrope is at $x_{\text{aniline}} = 0.0368$ and $T = 99.34$ °C.

6.7 Validation against independent data

We deliberately held the azeotropic data set AZEO001 out of the regression so it could serve as an independent check afterwards. Here’s how the comparison came out.

Quantity	Experimental	Model	Deviation
Azeotropic temperature	372.12 K (98.97 °C)	99.34 °C	0.37 K
Aniline mole fraction	0.04	0.0368	0.0032

This counts as a genuinely *independent* check, not just a self-consistency test — the regression never saw this azeotropic data. We fitted the model purely on liquid–liquid solubility data, and it still reproduced the mixture’s boiling behaviour to within 0.37 K. That said, the data set’s stated experimental uncertainty is 0.1 K, which puts us roughly four standard deviations off. The agreement is good, but it’s not within experimental error, and we’d rather say that plainly than oversell it as a perfect match.

6.8 Ternary behaviour

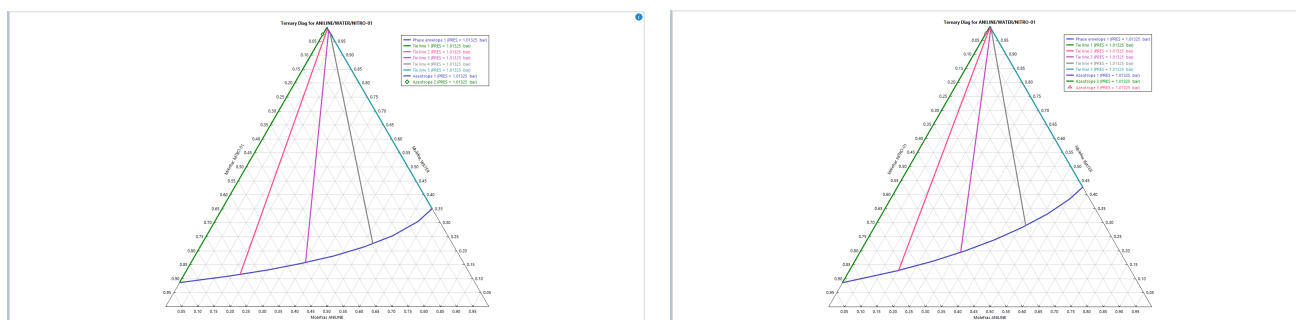


Figure 7: Ternary diagram for aniline/water/nitrobenzene, before regression (left) and after (right).

That wide two-liquid envelope tells us the decanter should give a clean split between an organic-rich phase and an aqueous phase across a broad range of feed compositions — good news for plant operability. Worth flagging too: a third azeotrope has shown up after the regression that wasn't there with the databank parameters. We'll need to pin that down before tackling the distillation sequencing in Report 3.

7 Objectives Which Could Not Be Accomplished, With Reasons

- **We haven't regressed the aniline–nitrobenzene pair yet.** The databank values here are thin too — A_{ij} and A_{ji} are both zero, only the B terms are present, and the validity range is a narrow 83–99 °C. We prioritised aniline–water first since the decanter depends on it directly; this pair is next on the list.

8 Any Other Challenges

- The regression run throws errors under “messages issued during Report Generation”. These come from the analysis calculations after the fit, at points where the model predicts a single liquid phase — not from the regression itself, so the parameters and their statistics are still valid.
- We have not yet settled on a single citable source for the LHHW kinetic parameters, so the reactor remains an `RStoic` block until Report 4.

9 Self-Assessment

The points on which we consider this report weakest are as follows.

1. The reactor performance is an assumption and not a result. The conversion and the selectivity split are specified by us on an `RStoic` block, and the material balance, heat duty and cost analysis all inherit that. *Kinetic model in Report 4.*
2. The gross contribution rests on three prices. The aniline realisation of Rs. 155,000/t sits marginally above the top of the observed band, and the nitrobenzene price assumes a term contract has actually been secured. *To be revisited with delivered quotations.*
3. Only the aniline–water pair has been regressed. Aniline–nitrobenzene is still on databank values valid over 83–99 °C, which is narrower than the aniline column will operate over, and the third azeotrope which appeared in the ternary diagram after regression has not yet been identified. *Both before the distillation sequencing of Report 3.*

10 Number of Hours Spent During This Period

Approximately **3 hours per member**, that is about **18 person-hours** for the team in this reporting period. This covers the literature survey, the hand material balance, the Aspen Plus laboratory session for the data retrieval and regression, and the preparation of this report.

11 Contributions From Individual Members

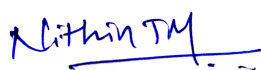
- **Nithin D H and Sajag Masane — process identification, specifications and block diagram.** Literature survey of the alternative synthesis routes and the comparison table, selection of the catalyst system and the reactor operating window, the reaction network including the two side reactions, the feed and product specifications, the hand material balance, and the block flow diagram with the three recycles.
- **Aditya Nitin Patil and Shubham Singh — input–output cost analysis and market analysis.** Market size and end-use survey, price and procurement basis for aniline, nitrobenzene, hydrogen and benzene, the consumption factors and the raw material cost build-up.
- **Nithin T M and Shivam Dongare — thermodynamic tuning, regression and plots.** Property method selection and the Henry component treatment, examination of the databank parameters and their validity limits, retrieval and screening of the NIST TDE data sets, the regression including the diagnosis of the over-parameterised four-parameter fit, the validation against the independent azeotropic data, and all the binary and ternary plots before and after regression.

All the six members have reviewed the complete report and each of us is able to explain the sections owned by the others.

12 References

1. T. Kahl et al., “Aniline”, *Ullmann’s Encyclopedia of Industrial Chemistry*, DOI 10.1002/14356007.a02.303.
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8. Price references, accessed August 2026: (a) aniline, India Q4 2025, 1500–1600 USD/MT, price-watch.ai; (b) nitrobenzene, India Q1 2026, 0.96 USD/kg, Expert Market Research; (c) benzene, India July 2026, 0.95 USD/kg, IMARC Group; (d) grey hydrogen production cost, India, 1.92–2.43 USD/kg, FIPI Hydrogen Market Report. Exchange rate USD 1 = Rs. 95 as on 6 August 2026.

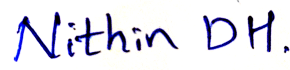
Signatures of members



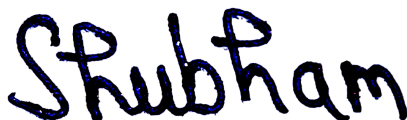
Nithin T M



Aditya Nitin Patil



Nithin D H



Shubham Singh



Shivam Dongare



Sajag Masane